

## Overview of project submission.

### 1. React

- **Why:** React's component-based architecture is ideal for managing UI states such as typing indicators, dark mode, and section toggling.
- **How:** Each part of the UI (Navbar, ChatInterface, Intro section) is encapsulated in its own component for reusability and isolation.

### 2. Tailwind CSS

- **Why:** Provides utility-first styling with responsive variants and theming capabilities.
- **How:** Enables rapid prototyping and dynamic styling such as `dark:bg-gray-900`, `transition-colors`, and responsiveness using `md:flex-row`.

### 3. Custom Hook (useTypewriter)

- **Why:** For animating the AI response text as if it's being typed in real-time.
- **How:** `useTypewriter(response, speed)` returns `text`, `isTyping`, and `stopTyping`, allowing fine control over animation and user interruption.

### 4. Context API

- **Why:** Theme switching (dark/light mode) needed to be globally accessible.
- **How:** `themeContext` provides `isDark` and `toggleTheme()` across the app via `useTheme()` hook.

### 5. SVG as Component

- **Why:** SVGs are scalable and customizable (e.g., adapt to theme color).
- **How:** from the genoshi website I copied the svg logo

## Few things which are implemented in this project from the source video given:

- The response text appears in typewriter like animation.
- When typing starts the response text goes a little less bright
- As typing starts the send button appears and as the text area goes blank it disappears
- When we press enter the text is sent.
- As response typing completes the textarea reappears

Extra features added:

- Dark and light theme toggle button
- Start conversation button and panel on left.
- New loader

How the script works:

- For the prototype sake I have added a function which when called by send button on click returns the same text given by the user.

Github link: <https://github.com/aditya-raj2230/chatinterface>

Live link: <https://chatinterface-five.vercel.app/>